20261020 Quiz

for (MATH230) Probability & Statistics

Jungwoo Kim
Department of Mathematics
Pohang University of Science and Technology

Exercise 5.55. The probability that a student pilot passes the written test for a private pilot's license is 0.7. Find the probability that a given student will pass the test

(a) on the third try;
(b) before the fourth try.

Solution. Assuming independent attempts with the same probability of passing, let $T \sim \text{Geom}(0.7)$ denote the trial on which the first pass occurs. Hence $\mathbb{P}(T = k) = 0.3^{k-1}(0.7)$.

(a) $\mathbb{P}(T = 3) = (0.3)^2(0.7) = \boxed{0.063}$,
(b) $\mathbb{P}(T < 4) = 1 - \mathbb{P}(T \geq 4) = 1 - (0.3)^3 = \boxed{0.973}$.

The second event allows success on any of the first three attempts.

□

Exercise 5.56. On average, 3 traffic accidents per month occur at a certain intersection. What is the probability that in any given month at this intersection

(a) exactly 5 accidents will occur?
(b) fewer than 3 accidents will occur?
(c) at least 2 accidents will occur?

Solution. Model the monthly number X of accidents by $X \sim \text{Pois}(3)$, so $\mathbb{P}(X = x) = e^{-3}3^x/x!$ for $x = 0, 1, \ldots$.

(a) $\mathbb{P}(X = 5) = \dfrac{e^{-3}3^5}{5!} \approx \boxed{0.1008}$,
(b) $\mathbb{P}(X < 3) = e^{-3}\left(1 + 3 + \dfrac{3^2}{2!}\right) \approx \boxed{0.4232}$,
(c) $\mathbb{P}(X \geq 2) = 1 - \mathbb{P}(X \leq 1) = 1 - e^{-3}(1 + 3) \approx \boxed{0.8009}$.

□

Exercise 5.57. On average, a textbook author makes two word-processing errors per page on the first draft of her textbook. What is the probability that on the next page she will make

(a) 4 or more errors?

(b) no errors?

Solution. Let $X \sim \text{Pois}(2)$ count errors on the next page.

$$\text{(a)} \quad \mathbb{P}(X \geq 4) = 1 - \mathbb{P}(X \leq 3) = 1 - e^{-2} \sum_{x=0}^{3} \frac{2^x}{x!} \approx \boxed{0.1429},$$

$$\text{(b)} \quad \mathbb{P}(X = 0) = e^{-2} \approx \boxed{0.1353}.$$

□

Exercise 5.62. The probability that a student at a local high school fails the screening test for scoliosis (curvature of the spine) is known to be 0.004. Of the next 1875 students at the school who are screened for scoliosis, find the probability that

(a) fewer than 5 fail the test;

(b) 8, 9, or 10 fail the test.

Solution. With independent student outcomes, the exact count is $X \sim \text{Bin}(1875, 0.004)$. Since n is large and p is small, use the Poisson approximation $Y \sim \text{Pois}(\mu)$ with

$$\mu = np = (1875)(0.004) = 7.5, \qquad \mathbb{P}(X = x) \approx \mathbb{P}(Y = x) = \frac{e^{-7.5}(7.5)^x}{x!}.$$

$$\text{(a)} \quad \mathbb{P}(X < 5) \approx \mathbb{P}(Y \leq 4) = e^{-7.5} \sum_{x=0}^{4} \frac{7.5^x}{x!} \approx \boxed{0.1321},$$

$$\text{(b)} \quad \mathbb{P}(X \in \{8, 9, 10\}) \approx \sum_{x=8}^{10} \frac{e^{-7.5} 7.5^x}{x!} \approx \boxed{0.3376}.$$

These are approximations, not exact binomial probabilities.

□

Exercise 5.66. Changes in airport procedures require considerable planning. Arrival rates of aircraft are important factors that must be taken into account. Suppose small aircraft arrive at a certain airport, according to a Poisson process, at the rate of 6 per hour. Thus, the Poisson parameter for arrivals over a period of t hours is $\mu = 6t$.

(a) What is the probability that exactly 4 small aircraft arrive during a 1-hour period?

(b) What is the probability that at least 4 arrive during a 1-hour period?

(c) If we define a working day as 12 hours, what is the probability that at least 75 small aircraft arrive during a working day?

Solution. For a period of length t hours, let $N_t \sim \text{Pois}(6t)$.

$$\text{(a)} \quad \mathbb{P}(N_1 = 4) = \frac{e^{-6}6^4}{4!} \approx \boxed{0.1339},$$

$$\text{(b)} \quad \mathbb{P}(N_1 \geq 4) = 1 - \mathbb{P}(N_1 \leq 3) = 1 - e^{-6} \sum_{x=0}^{3} \frac{6^x}{x!} \approx \boxed{0.8488}.$$

(c) Since 12 hours have mean $6(12) = 72$ arrivals,

$$\mathbb{P}(N_{12} \geq 75) = 1 - \mathbb{P}(N_{12} \leq 74) = 1 - e^{-72} \sum_{x=0}^{74} \frac{72^x}{x!} \approx \boxed{0.3773}.$$

The large sum may be evaluated from a cumulative Poisson table or software.

□

Exercise 5.71. For a certain type of copper wire, it is known that, on average, 1.2 flaws occur per millimeter. Assuming that the number of flaws is a Poisson random variable, what is the probability that no flaws occur in a certain portion of wire of length 5 millimeters? What is the mean number of flaws in a portion of length 5 millimeters?

Solution. The expected count over 5 millimeters is $\mu = (1.2)(5) = 6$. If $X \sim \text{Pois}(6)$ counts the flaws in that portion, then

$$\boxed{\mathbb{P}(X = 0) = e^{-6} \approx 0.0025}, \qquad \boxed{\mathbb{E}[X] = 6}.$$

□

Exercise 5.73. Hospital administrators in large cities anguish about traffic in emergency rooms. At a particular hospital in a large city, the staff on hand cannot accommodate the patient traffic if there are more than 10 emergency cases in a given hour. It is assumed that patient arrival follows a Poisson process, and historical data suggest that, on the average, 5 emergencies arrive per hour.

(a) What is the probability that in a given hour the staff cannot accommodate the patient traffic?

(b) What is the probability that more than 20 emergencies arrive during a 3-hour shift?

Solution. For t hours, let $N_t \sim \text{Pois}(5t)$.

$$\text{(a)} \quad \mathbb{P}(N_1 > 10) = 1 - e^{-5} \sum_{x=0}^{10} \frac{5^x}{x!} \approx \boxed{0.0137},$$

$$\text{(b)} \quad \mathbb{P}(N_3 > 20) = 1 - e^{-15} \sum_{x=0}^{20} \frac{15^x}{x!} \approx \boxed{0.0830}.$$

The 3-hour mean is $5(3) = 15$, rather than 5.

□

Exercise 5.74. It is known that 2% of people whose luggage is screened at an airport have questionable objects in their luggage. What is the probability that a string of 15 people pass through screening successfully before an individual is caught with a questionable object? What is the expected number of people to pass through before an individual is stopped?

Solution. Assuming independent screenings, let $T \sim \text{Geom}(0.02)$ be the screening number of the first individual caught, and let $Y = T - 1$ count people who pass before that individual. The specified sequence is 15 successful screenings followed by one detection, so

$$\boxed{\mathbb{P}(Y = 15) = (0.98)^{15}(0.02) \approx 0.0148}.$$

Since $\mathbb{E}[T] = 1/0.02 = 50$,

$$\boxed{\mathbb{E}[Y] = \mathbb{E}[T] - 1 = 50 - 1 = 49 \text{ people}}.$$

□

Exercise 5.76. The refusal rate for telephone polls is known to be approximately 20%. A newspaper report indicates that 50 people were interviewed before the first refusal.

(a) Comment on the validity of the report. Use a probability in your argument.

(b) What is the expected number of people interviewed before a refusal?

Solution. Assume independent responses with refusal probability $p = 0.20$. Let $T \sim \text{Geom}(0.20)$ be the position of the first refusal, and $Y = T - 1$ the number of people interviewed before it.

(a) The stated event consists of 50 interviews followed by a refusal:

$$\boxed{\mathbb{P}(Y = 50) = (0.80)^{50}(0.20) \approx 2.85 \times 10^{-6}}.$$

This is extraordinarily unlikely under the assumed refusal rate, so the report is questionable, although not logically impossible.

(b) The mean number interviewed before the first refusal is

$$\boxed{\mathbb{E}[Y] = \mathbb{E}[T] - 1 = \frac{1}{0.20} - 1 = 4}.$$

□

Exercise 5.82. An electronic switching device occasionally malfunctions, but the device is considered satisfactory if it makes, on average, no more than 0.20 error per hour. A particular 5-hour period is chosen for testing the device. If no more than 1 error occurs during the time period, the device will be considered satisfactory.

(a) What is the probability that a satisfactory device will be considered unsatisfactory on the basis of the test? Assume a Poisson process.

(b) What is the probability that a device will be accepted as satisfactory when, in fact, the mean number of errors is 0.25? Again, assume a Poisson process.

Solution. Let X count errors in 5 hours. The device passes the inspection if $X \le 1$ and fails if $X \ge 2$.

(a) At the satisfactory boundary rate 0.20 error per hour, $X \sim \text{Pois}(5(0.20)) = \text{Pois}(1)$. Thus

$$\boxed{\mathbb{P}(X \ge 2) = 1 - \mathbb{P}(X \le 1) = 1 - e^{-1}(1+1) \approx 0.2642}.$$

For a lower true error rate this false-rejection probability would be smaller.

(b) At the unsatisfactory rate 0.25 error per hour, $X \sim \text{Pois}(5(0.25)) = \text{Pois}(1.25)$. Its erroneous-acceptance probability is

$$\boxed{\mathbb{P}(X \le 1) = e^{-1.25}(1 + 1.25) \approx 0.6446}.$$

□

Exercise 5.93. It is known by researchers that 2 in 100 people carry a gene that leads to the inheritance of a certain chronic disease. In a random sample of 1000 individuals, what is the probability that fewer than 11 individuals carry the gene? Use a Poisson approximation. Again, using the approximation, what is the approximate mean number of people out of 1000 carrying the gene?

Solution. Under independent sampling, the exact count has distribution $X \sim \text{Bin}(1000, 0.02)$. The Poisson approximation is $Y \sim \text{Pois}(np) = \text{Pois}(20)$. Hence

$$\boxed{\mathbb{P}(X < 11) \approx \mathbb{P}(Y \le 10) = e^{-20} \sum_{x=0}^{10} \frac{20^x}{x!} \approx 0.0108}.$$

The approximate mean count is

$$\boxed{\mathbb{E}[Y] = np = (1000)(0.02) = 20 \text{ people}}.$$

□

Exercise 6.1. Given a continuous uniform distribution on $[A, B]$ with $A < B$, show that

(a) $\mu = \dfrac{A+B}{2}$;

(b) $\sigma^2 = \dfrac{(B-A)^2}{12}$.

Solution. For $X \sim \text{Unif}(A, B)$, the density is $f_X(x) = 1/(B-A)$ for $A \le x \le B$, and zero otherwise.

(a) Integrating against the density gives

$$\boxed{\mu = \mathbb{E}[X] = \int_A^B \frac{x}{B-A}\, \mathrm{d}x = \frac{B^2 - A^2}{2(B-A)} = \frac{A+B}{2}}.$$

(b) First calculate the second moment:

$$\mathbb{E}[X^2] = \int_A^B \frac{x^2}{B-A}\, \mathrm{d}x = \frac{B^3 - A^3}{3(B-A)} = \frac{A^2 + AB + B^2}{3}.$$

Consequently,

$$\begin{aligned} \sigma^2 = \text{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\ &= \frac{A^2 + AB + B^2}{3} - \frac{(A+B)^2}{4} = \boxed{\frac{(B-A)^2}{12}}. \end{aligned}$$

□

Exercise 6.2. Suppose X follows a continuous uniform distribution from 0 to 5. Determine the conditional probability $\mathbb{P}(X < 3.5 \mid X \geq 1)$.

Solution. For $X \sim \text{Unif}(0, 5)$, the event $X < 3.5$ intersected with $X \geq 1$ is the interval $[1, 3.5)$. Therefore

$$\begin{aligned}\mathbb{P}(X < 3.5 \mid X \geq 1) &= \frac{\mathbb{P}(1 \leq X < 3.5)}{\mathbb{P}(X \geq 1)} \\ &= \frac{(3.5-1)/5}{(5-1)/5} = \boxed{\frac{5}{8} = 0.625}.\end{aligned}$$

□

Exercise 6.3. The daily milk consumption, in liters, dispensed by a milk booth located in a city is a random variable X, having a continuous uniform distribution with $A = 90$ and $B = 120$. Find the probability that on a given day, the amount of milk dispensed by this booth will be

(a) at most 100 liters;
(b) more than 105 liters but less than 115 liters;
(c) at least 105 liters.

Solution. Since $X \sim \text{Unif}(90, 120)$, its probability density is $1/30$ on $[90, 120]$; the probability of an interval equals its length divided by 30.

$$\text{(a)} \quad \mathbb{P}(X \leq 100) = \frac{100-90}{120-90} = \boxed{\frac{1}{3}},$$

$$\text{(b)} \quad \mathbb{P}(105 < X < 115) = \frac{115-105}{120-90} = \boxed{\frac{1}{3}},$$

$$\text{(c)} \quad \mathbb{P}(X \geq 105) = \frac{120-105}{120-90} = \boxed{\frac{1}{2}}.$$

Endpoints have probability zero under a continuous distribution.

□

Exercise 6.4. A train arrives at a station every 15 minutes. It is assumed that the waiting time for a particular individual is a random variable with a continuous uniform distribution.

(a) What is the probability that the individual waits more than 10 minutes?
(b) What is the probability that the individual waits between 6 and 12 minutes?

Solution. The waiting time $X \sim \text{Unif}(0, 15)$ is uniformly distributed within one train interval.

$$\text{(a)} \quad \mathbb{P}(X > 10) = \frac{15-10}{15} = \boxed{\frac{1}{3}},$$

$$\text{(b)} \quad \mathbb{P}(6 < X < 12) = \frac{12-6}{15} = \boxed{\frac{2}{5}}.$$

□

Exercise 6.5. Given a standard normal distribution, find the area under the curve that lies

(a) to the left of $z = -1.10$;
(b) to the right of $z = 1.645$;
(c) between $z = -2.43$ and $z = -0.45$;
(d) to the left of $z = 0.45$;
(e) to the right of $z = -0.45$;
(f) between $z = -0.45$ and $z = 0.45$.

Solution. Let $Z \sim \mathcal{N}(0,1)$ and write $\Phi(z) = \mathbb{P}(Z \le z)$. By complementarity and symmetry,

$$\begin{aligned}
\text{(a)}\quad \mathbb{P}(Z < -1.10) &= \Phi(-1.10) \approx \boxed{0.1357},\\
\text{(b)}\quad \mathbb{P}(Z > 1.645) &= 1 - \Phi(1.645) \approx \boxed{0.0500},\\
\text{(c)}\quad \mathbb{P}(-2.43 < Z < -0.45) &= \Phi(-0.45) - \Phi(-2.43) \approx \boxed{0.3188},\\
\text{(d)}\quad \mathbb{P}(Z < 0.45) &= \Phi(0.45) \approx \boxed{0.6736},\\
\text{(e)}\quad \mathbb{P}(Z > -0.45) &= 1 - \Phi(-0.45) = \Phi(0.45) \approx \boxed{0.6736},\\
\text{(f)}\quad \mathbb{P}(-0.45 < Z < 0.45) &= 2\Phi(0.45) - 1 \approx \boxed{0.3473}.
\end{aligned}$$

The displayed decimals use the normal CDF at full precision before rounding; summing rounded table entries can differ in the last digit.

□

Exercise 6.6. Find the value of z if the area under a standard normal curve

(a) to the right of z is 0.3745;
(b) to the left of z is 0.3050;
(c) between $-z$ and 0, with $-z < 0$, is 0.4838;
(d) between $-z$ and z, with $z > 0$, is 0.90.

Solution. Use $\Phi(z) = \mathbb{P}(Z \le z)$ for $Z \sim \mathcal{N}(0,1)$, together with $\Phi(-z) = 1 - \Phi(z)$.

$$\begin{aligned}
\text{(a)}\quad \Phi(z) = 1 - 0.3745 = 0.6255 &\implies \boxed{z \approx 0.32},\\
\text{(b)}\quad \Phi(z) = 0.3050 &\implies \boxed{z \approx -0.51},\\
\text{(c)}\quad \Phi(0) - \Phi(-z) = 0.4838 &\implies \Phi(-z) = 0.0162,\ \boxed{z \approx 2.14},\\
\text{(d)}\quad \Phi(z) - \Phi(-z) = 0.90 &\implies \Phi(z) = 0.95,\ \boxed{z \approx 1.645}.
\end{aligned}$$

In (c), the required left endpoint $-z$ is negative; in (d), the central area leaves 0.05 in each tail.

□

Exercise 6.7. Given the standard normal distribution, find the value of k such that

(a) $\mathbb{P}(Z > k) = 0.9625$;
(b) $\mathbb{P}(Z < k) = 0.6255$;
(c) $\mathbb{P}(0.17 < Z < k) = 0.367$.

Solution. With $Z \sim \mathcal{N}(0,1)$ and $\Phi(k) = \mathbb{P}(Z \le k)$,

$$\begin{aligned}
\text{(a)}\quad \Phi(k) = 1 - 0.9625 = 0.0375 &\implies \boxed{k \approx -1.78},\\
\text{(b)}\quad \Phi(k) = 0.6255 &\implies \boxed{k \approx 0.32},\\
\text{(c)}\quad \Phi(k) = \Phi(0.17) + 0.367 \approx 0.5675 + 0.367 = 0.9345 &\implies \boxed{k \approx 1.51}.
\end{aligned}$$

□

Exercise 6.8. Given a normal distribution with $\mu = 30$ and $\sigma = 6$, find

(a) the normal curve area to the right of $x = 17$;
(b) the normal curve area to the left of $x = 22$;
(c) the normal curve area between $x = 32$ and $x = 41$;
(d) the value of x that has 80% of the normal curve area to the left;
(e) the two values of x that contain the middle 75% of the normal curve area.

Solution. Let $X \sim \mathcal{N}(30, 6^2)$, so $Z = (X - 30)/6 \sim \mathcal{N}(0, 1)$. Use $\Phi(z) = \mathbb{P}(Z \le z)$.

(a) Since $(17 - 30)/6 = -13/6$,

$$\boxed{\mathbb{P}(X > 17) = 1 - \Phi\left(-\frac{13}{6}\right) \approx 0.9849}.$$

(b) Since $(22 - 30)/6 = -4/3$,

$$\boxed{\mathbb{P}(X < 22) = \Phi\left(-\frac{4}{3}\right) \approx 0.0912}.$$

(c) The standardized endpoints are $(32 - 30)/6 = 1/3$ and $(41 - 30)/6 = 11/6$, so

$$\boxed{\mathbb{P}(32 < X < 41) = \Phi\left(\frac{11}{6}\right) - \Phi\left(\frac{1}{3}\right) \approx 0.3361}.$$

(d) The 80th percentile of the standard normal is $z_{0.80} \approx 0.8416$. Thus

$$\boxed{x = 30 + 6z_{0.80} \approx 35.05}.$$

(e) The middle 75% leaves 0.125 in each tail, so its endpoints correspond to $z_{0.125} \approx -1.1503$ and $z_{0.875} \approx 1.1503$. Hence

$$\boxed{x_1 = 30 + 6z_{0.125} \approx 23.10}, \qquad \boxed{x_2 = 30 + 6z_{0.875} \approx 36.90}.$$

□

Exercise 6.10. According to Chebyshev's theorem, the probability that any random variable assumes a value within 2 standard deviations of the mean is at least 0.75. If it is known that the probability distribution of a random variable X is normal, with mean μ and variance σ^2, what is the exact value of $\mathbb{P}(\mu - 2\sigma < X < \mu + 2\sigma)$?

Solution. Standardize using $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$. The exact probability can be expressed in terms of the standard normal CDF:

$$\begin{aligned}\mathbb{P}(\mu - 2\sigma < X < \mu + 2\sigma) &= \mathbb{P}(-2 < Z < 2) \\ &= \boxed{\Phi(2) - \Phi(-2) = 2\Phi(2) - 1} \approx \boxed{0.9545}.\end{aligned}$$

Chebyshev's 0.75 is a general lower bound, not the normal probability.

□